CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 9: MongoDB Document Modeling & Aggregation Pipelines

Course: Advanced Database Engineering Laboratory [23CS3506]
Category: Experiments / Practicals | Experiment #9 | Academic Year: 2026-27

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1. Overview & Objective:

Design non-relational document collections, execute multi-stage aggregation pipelines.

2. Practical Tasks & Verification Steps:

1. Model university courses document collection with embedded instructor and review arrays in MongoDB.
2. Execute insertMany inserting 10 sample course documents into collection.
3. Build multi-stage aggregation pipeline using $match department , $unwind reviews , $group average rating .
4. Export aggregation results sorted in descending order of average student rating using $sort and $project.

Submission Guidelines & Verification:

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